

# Virtual LAN (VLAN)

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INFORMATIKA UNS – MANAJEMEN JARINGAN GASAL 2026/2027

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# Apa itu VLAN?

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# Apa itu VLAN?

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- LANs are not necessarily restricted in size. A LAN can have 20 computers, 200 computers, or even more.
- The number of computers in the LAN is an issue, because **Layer 2** switches **do not separate broadcast domains**. This means that any broadcast sent out on the network (for example, the broadcast associated with an ARP request) will be **sent to all computers** in the LAN.
- **Excessive broadcasts** are a problem, because each computer must process the broadcast to determine whether it needs to respond; this essentially **slows down the computer and the network**.

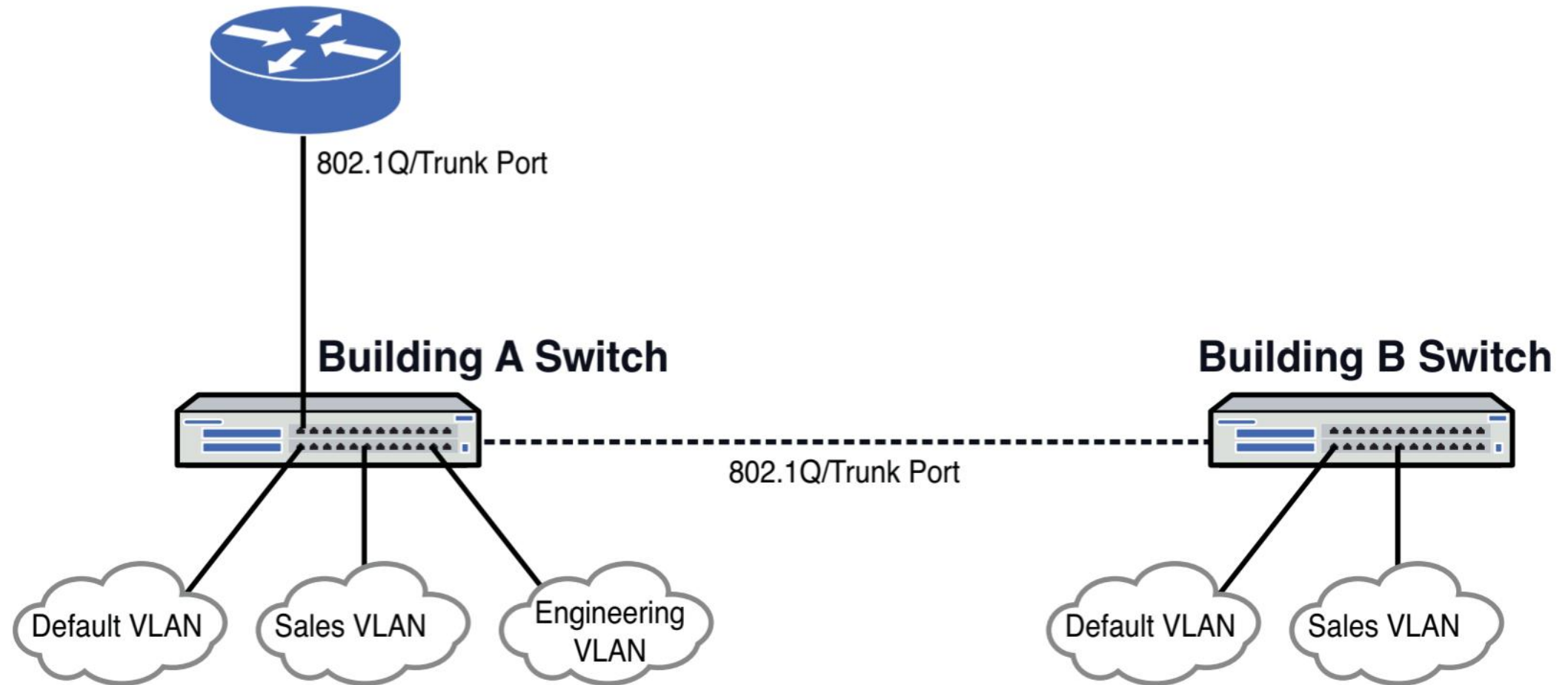
# Apa itu VLAN?

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- A network with multiple LANs interconnected at the **Layer 2** level is called a **flat network**. A flat network is where the LANs share the same broadcast domain.
- Flat networks can be avoided by the use of **virtual LANs (VLAN)** or **routers**. Although both options can be used to separate broadcast domains, they differ in that the **VLAN** operates at **the OSI Layer 2**, while **routers** use **Layer 3** networking to accomplish the task.
- A **VLAN** is a way to have multiple LANs co-exist in the same Layer 2 switch, but their traffic is segregated from each other. Even though they reside on the same physical switch, they behave as if they are on different switches (hence, the term **virtual**).

# Apa Itu VLAN?

- Contoh VLAN:

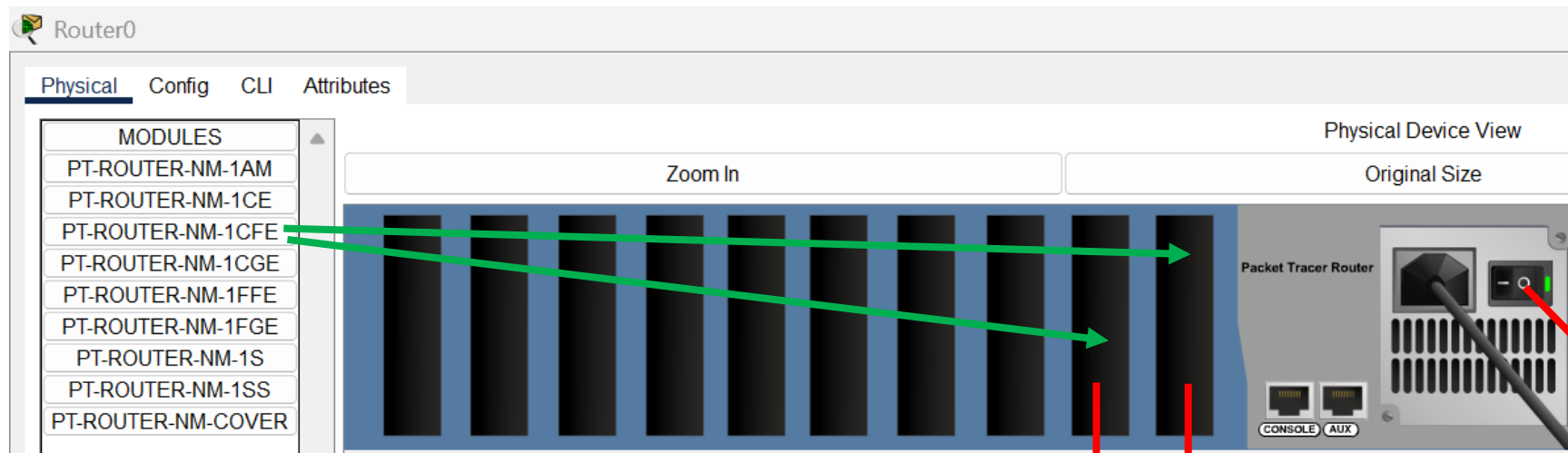


# VLAN dengan Cisco Packet Tracer

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# VLAN dengan Cisco Packet Tracer

- Switch: 2960
- Router: PT-Empty (supaya bisa di-custom jumlah interfacenya)
  - Contoh, cara untuk menambahkan 2 buah interface FastEthernet (Fa):

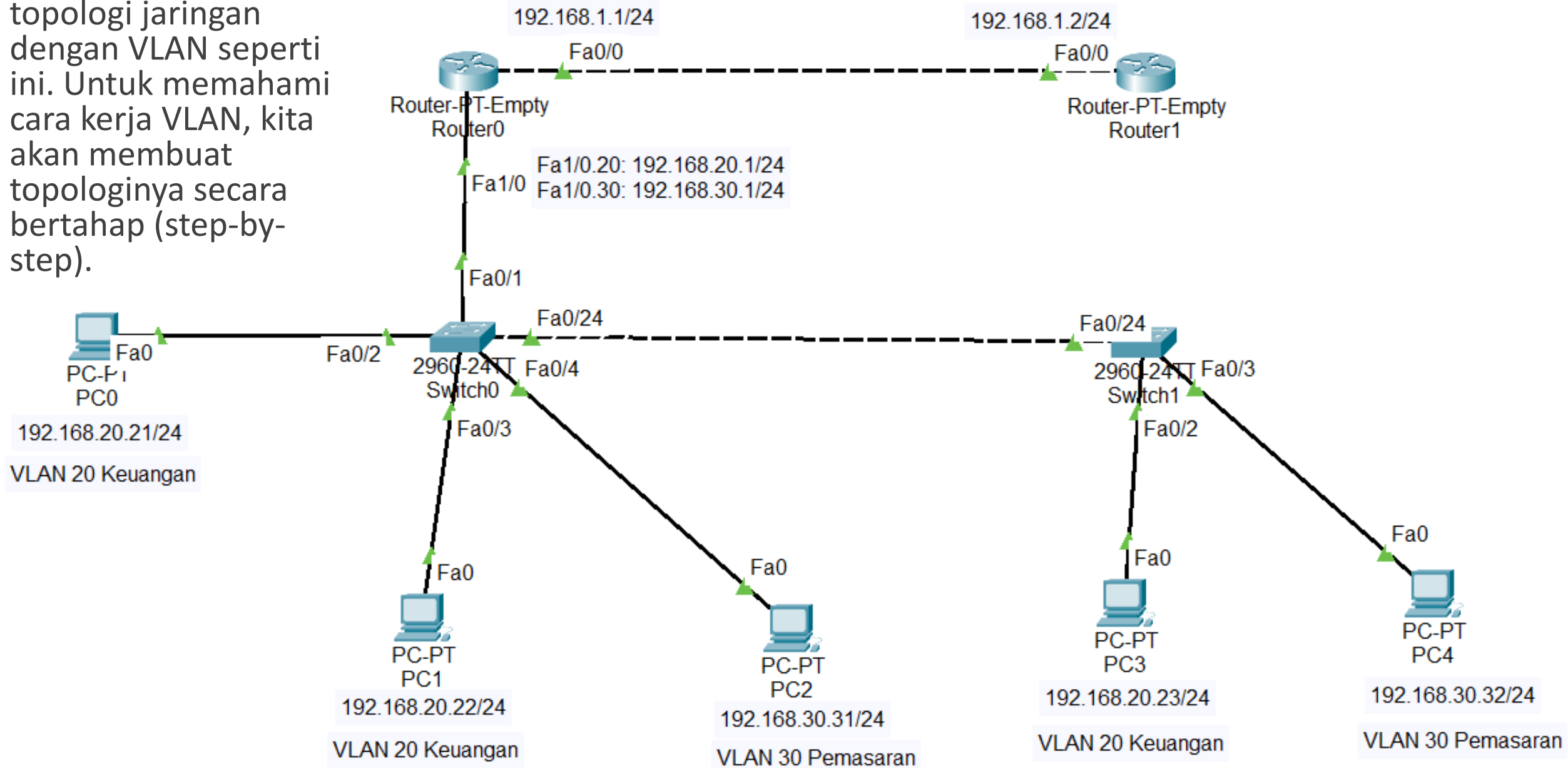


1. Matikan router dulu.

2. Yang **PT-ROUTER-NM-1CFE** geser aja kesini 2 buah (dari kanan dulu).

# VLAN dengan Cisco Packet Tracer

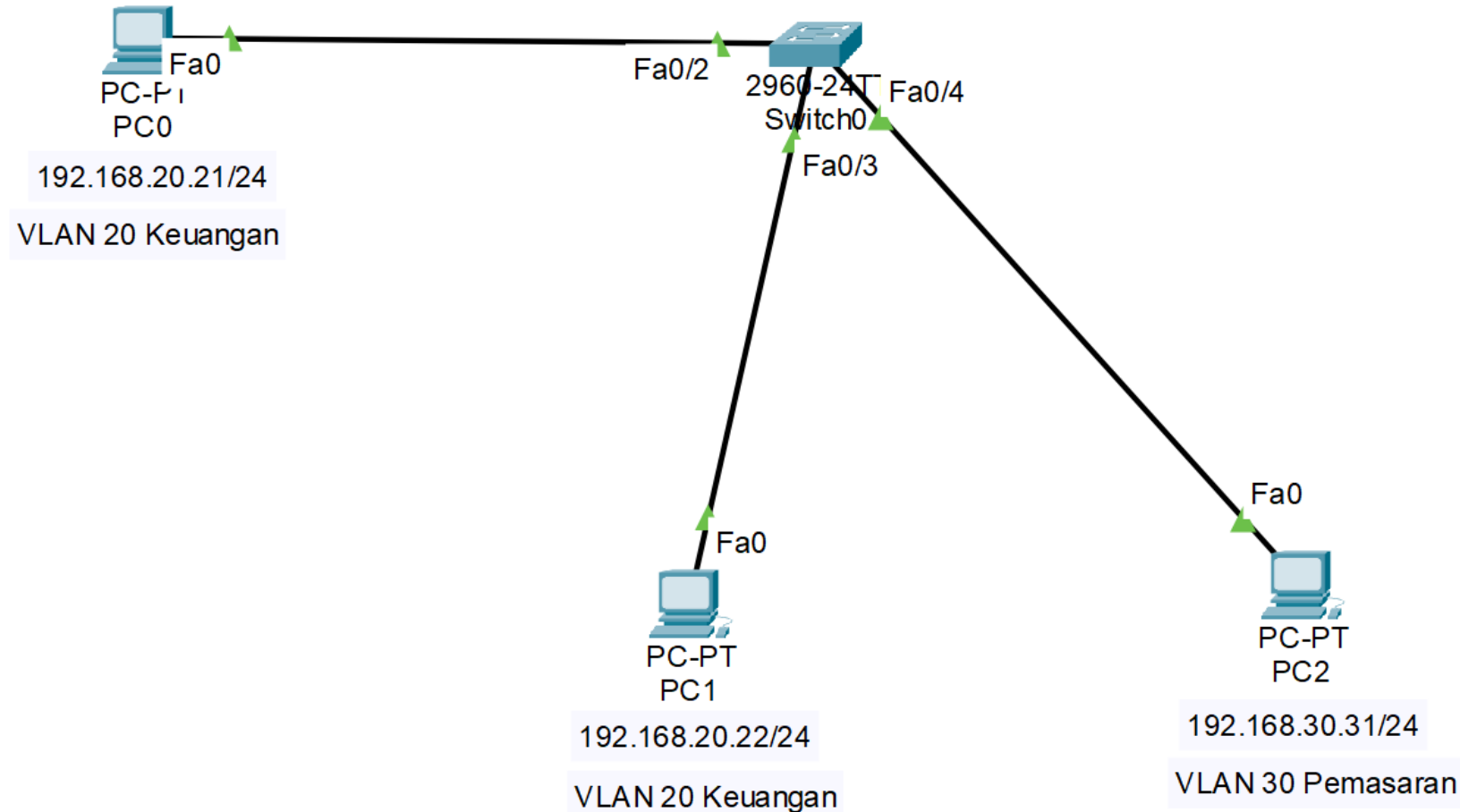
Kita akan membuat topologi jaringan dengan VLAN seperti ini. Untuk memahami cara kerja VLAN, kita akan membuat topologinya secara bertahap (step-by-step).



# VLAN dengan Cisco Packet Tracer

## Ping antar PC tanpa default gateway

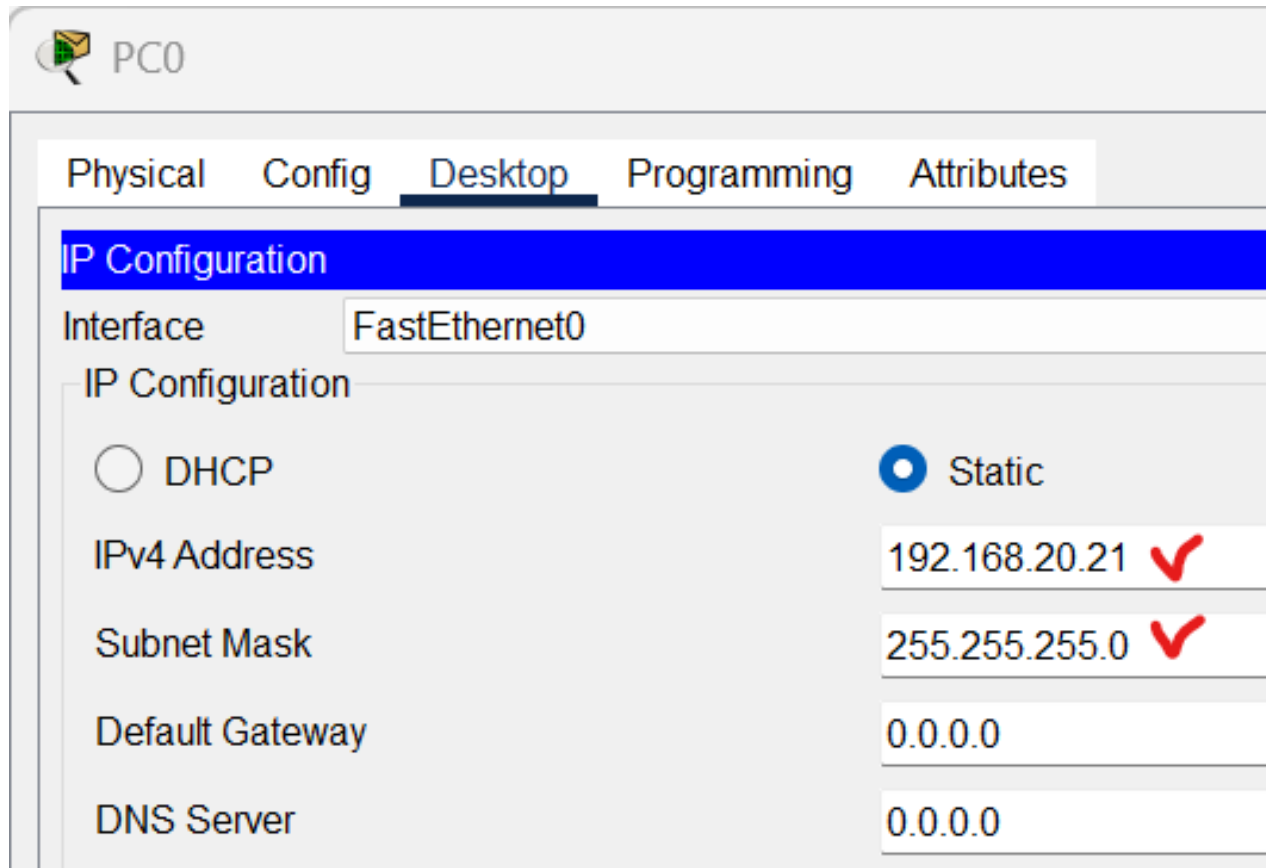
- Kita buat dulu topologi dengan Switch0 dan 3 PC seperti ini:



# VLAN dengan Cisco Packet Tracer

## Ping antar PC tanpa default gateway

- Isi semua konfigurasi IP PC nya **tanpa default gateway**. Contoh pada PC0:



The screenshot shows the configuration window for PC0 in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'IP Configuration' section is highlighted. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is set to 192.168.20.21, the Subnet Mask is 255.255.255.0, the Default Gateway is 0.0.0.0, and the DNS Server is 0.0.0.0. Red checkmarks are visible next to the IP address and subnet mask fields.

| Interface                  | FastEthernet0                           |
|----------------------------|-----------------------------------------|
| IP Configuration           |                                         |
| <input type="radio"/> DHCP | <input checked="" type="radio"/> Static |
| IPv4 Address               | 192.168.20.21 ✓                         |
| Subnet Mask                | 255.255.255.0 ✓                         |
| Default Gateway            | 0.0.0.0                                 |
| DNS Server                 | 0.0.0.0                                 |

# VLAN dengan Cisco Packet Tracer

## Ping antar PC tanpa default gateway

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- Coba ping dari PC0 ke PC1, sukses:

```
C:\>ping 192.168.20.22

Pinging 192.168.20.22 with 32 bytes of data:

Reply from 192.168.20.22: bytes=32 time=11ms TTL=128
Reply from 192.168.20.22: bytes=32 time<1ms TTL=128
Reply from 192.168.20.22: bytes=32 time<1ms TTL=128
Reply from 192.168.20.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

- Coba ping dari PC0 ke PC2, tidak sukses:

```
C:\>ping 192.168.30.31

Pinging 192.168.30.31 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.31:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

Konfigurasikan perintah berikut pada Switch0:

- Switch>enable
- Switch#show vlan [\[melihat daftar VLAN\]](#)
  - Secara default sudah ada VLAN dengan id 1 beserta port nya:

| VLAN | Name               | Status | Ports                                                                                                                                                                                                             |
|------|--------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2 |
| 1002 | fddi-default       | active |                                                                                                                                                                                                                   |
| 1003 | token-ring-default | active |                                                                                                                                                                                                                   |
| 1004 | fddinet-default    | active |                                                                                                                                                                                                                   |
| 1005 | trnet-default      | active |                                                                                                                                                                                                                   |

| VLAN | Type  | SAID   | MTU  | Parent | RingNo | BridgeNo | Stp  | BrdgMode | Trans1 | Trans2 |
|------|-------|--------|------|--------|--------|----------|------|----------|--------|--------|
| 1    | enet  | 100001 | 1500 | -      | -      | -        | -    | -        | 0      | 0      |
| 1002 | fddi  | 101002 | 1500 | -      | -      | -        | -    | -        | 0      | 0      |
| 1003 | tr    | 101003 | 1500 | -      | -      | -        | -    | -        | 0      | 0      |
| 1004 | fdnet | 101004 | 1500 | -      | -      | -        | ieee | -        | 0      | 0      |
| 1005 | trnet | 101005 | 1500 | -      | -      | -        | ibm  | -        | 0      | 0      |

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

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Menambahkan port pada VLAN 20 dan VLAN 30:

- Switch#configure terminal
- Switch(config)#vlan 20 [menambah VLAN dengan id: 20]
- Switch(config-vlan)#name Keuangan [menamai VLAN id 20 dengan: Keuangan]
- Switch(config-vlan)#vlan 30
- Switch(config-vlan)#name Pemasaran
- Switch(config-vlan)#exit
- Switch(config)#exit
  
- Kalau VLAN tidak dinamai, perintah dengan warna hijau di atas bisa di-skip. Bisa langsung pakai Switch(config-if)#switchport access vlan <<id\_vlan>> saat setting di interface nya.
- Jika ingin menghapus VLAN, gunakan perintah:
  - Switch(config)#no vlan <<id\_vlan>> [menghapus VLAN dengan id tertentu]

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

- Switch#show vlan

- Terlihat sudah ada VLAN **Keuangan** dan **Pemasaran**, tapi belum ada port-port nya.

```
Switch#show vlan
```

| VLAN Name |                    | Status | Ports                                                                                                                                                                                                             |
|-----------|--------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -----     |                    |        |                                                                                                                                                                                                                   |
| 1         | default            | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2 |
| 20        | Keuangan           | active |                                                                                                                                                                                                                   |
| 30        | Pemasaran          | active |                                                                                                                                                                                                                   |
| 1002      | fddi-default       | active |                                                                                                                                                                                                                   |
| 1003      | token-ring-default | active |                                                                                                                                                                                                                   |
| 1004      | fddinet-default    | active |                                                                                                                                                                                                                   |
| 1005      | trnet-default      | active |                                                                                                                                                                                                                   |

| VLAN  | Type | SAID   | MTU  | Parent | RingNo | BridgeNo | Stp | BrdgMode | Trans1 | Trans2 |
|-------|------|--------|------|--------|--------|----------|-----|----------|--------|--------|
| ----- |      |        |      |        |        |          |     |          |        |        |
| 1     | enet | 100001 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 20    | enet | 100020 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 30    | enet | 100030 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |

```
--More-- |
```

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

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Menambahkan port pada VLAN 20 dan VLAN 30:

- Switch#configure terminal
- Switch(config)#interface fastEthernet 0/2 [masuk interface Fa0/2 pada switch]
- Switch(config-if)#switchport mode access [access: port ke end device]
- Switch(config-if)#switchport access vlan 20
- Switch(config-if)#exit
- Switch(config)#interface fastEthernet 0/3
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 20
- Switch(config-if)#exit

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

---

- Switch(config)#interface fastEthernet 0/4
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 30
- Switch(config-if)#exit
- Switch(config)#exit

# VLAN dengan Cisco Packet Tracer

## Switch0 – Menambahkan VLAN

- Switch#show vlan
- Terlihat VLAN **Keuangan** dan **Pemasaran** sudah ada port nya :

| VLAN | Name               | Status | Ports                                                                                                                                                                                      |
|------|--------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/5, Fa0/6, Fa0/7<br>Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15<br>Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24, Gig0/1, Gig0/2 |
| 20   | Keuangan           | active | Fa0/2, Fa0/3                                                                                                                                                                               |
| 30   | Pemasaran          | active | Fa0/4                                                                                                                                                                                      |
| 1002 | fddi-default       | active |                                                                                                                                                                                            |
| 1003 | token-ring-default | active |                                                                                                                                                                                            |
| 1004 | fddinet-default    | active |                                                                                                                                                                                            |
| 1005 | trnet-default      | active |                                                                                                                                                                                            |

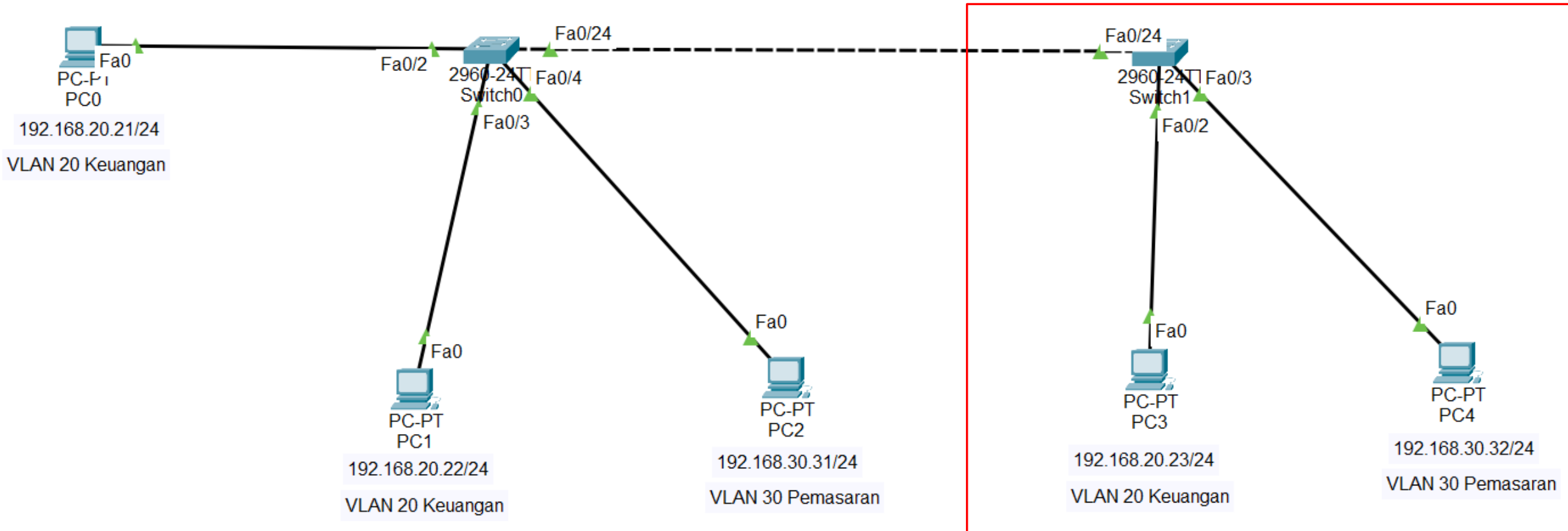
| VLAN | Type | SAID   | MTU  | Parent | RingNo | BridgeNo | Stp | BrdgMode | Trans1 | Trans2 |
|------|------|--------|------|--------|--------|----------|-----|----------|--------|--------|
| 1    | enet | 100001 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 20   | enet | 100020 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 30   | enet | 100030 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 1002 | fddi | 101002 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |

--More-- |

# VLAN dengan Cisco Packet Tracer

## Tambahkan Switch1, PC3, dan PC4

- Tambahkan topologi seperti yang dikotakin di sebelah kanan. Lalu isi semua konfigurasi IP pada PC3 dan PC4 **tanpa default gateway**.



# VLAN dengan Cisco Packet Tracer

## Switch1 – Menambahkan VLAN

Menambahkan VLAN 20 dan VLAN 30 beserta port nya (VLAN tidak dinamai):

- Switch>enable
- Switch#configure terminal
- Switch(config)#interface fastEthernet 0/2
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 20 [bisa langsung menambahkan vlan dengan id 20 tanpa menamainya] 

% Access VLAN does not exist. Creating vlan 20
- Switch(config-if)#exit
- Switch(config)#interface fastEthernet 0/3
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 30
- Switch(config-if)#exit
- Switch(config)#exit

# VLAN dengan Cisco Packet Tracer

## Switch1 – Menambahkan VLAN

- Switch#show vlan
- Terlihat ada VLAN Keuangan dan Pemasaran beserta port nya, tapi **tidak ada nama VLAN** nya:

| VLAN | Name               | Status | Ports                                                                                                                                                                                             |
|------|--------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/4, Fa0/5, Fa0/6<br>Fa0/7, Fa0/8, Fa0/9, Fa0/10<br>Fa0/11, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/18<br>Fa0/19, Fa0/20, Fa0/21, Fa0/22<br>Fa0/23, Fa0/24, Gig0/1, Gig0/2 |
| 20   | VLAN0020           | active | Fa0/2                                                                                                                                                                                             |
| 30   | VLAN0030           | active | Fa0/3                                                                                                                                                                                             |
| 1002 | fddi-default       | active |                                                                                                                                                                                                   |
| 1003 | token-ring-default | active |                                                                                                                                                                                                   |
| 1004 | fddinet-default    | active |                                                                                                                                                                                                   |
| 1005 | trnet-default      | active |                                                                                                                                                                                                   |

| VLAN | Type | SAID   | MTU  | Parent | RingNo | BridgeNo | Stp | BrdgMode | Trans1 | Trans2 |
|------|------|--------|------|--------|--------|----------|-----|----------|--------|--------|
| 1    | enet | 100001 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 20   | enet | 100020 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 30   | enet | 100030 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |
| 1002 | fddi | 101002 | 1500 | -      | -      | -        | -   | -        | 0      | 0      |

--More--

# VLAN dengan Cisco Packet Tracer

## Trunk Switch0 ke Switch1

---

- Coba ping dari PC0 ke PC3, belum sukses:

```
C:\>ping 192.168.20.23

Pinging 192.168.20.23 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.23:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Biar bisa ping dari PC0 ke PC3, kita perlu “menyambungkan” Switch0 dan Switch1 yang kedua switch tadi telah dikonfigurasi VLAN. Caranya dengan trunk.

# VLAN dengan Cisco Packet Tracer

## Trunk Switch0 ke Switch1

---

Konfigurasi perintah berikut pada **Switch0** dan **Switch1**:

- Switch>enable
- Switch#configure terminal
- Switch(config)#interface fastEthernet 0/24
- Switch(config-if)#switchport mode **trunk** [**trunk: port ke switch/router**]

# VLAN dengan Cisco Packet Tracer

## Trunk Switch0 ke Switch1

- Coba ping dari PC0 ke PC3, sudah sukses:

```
C:\>ping 192.168.20.23

Pinging 192.168.20.23 with 32 bytes of data:

Reply from 192.168.20.23: bytes=32 time<1ms TTL=128
Reply from 192.168.20.23: bytes=32 time=17ms TTL=128
Reply from 192.168.20.23: bytes=32 time<1ms TTL=128
Reply from 192.168.20.23: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.23:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

- Coba ping dari PC0 ke PC4, belum sukses:

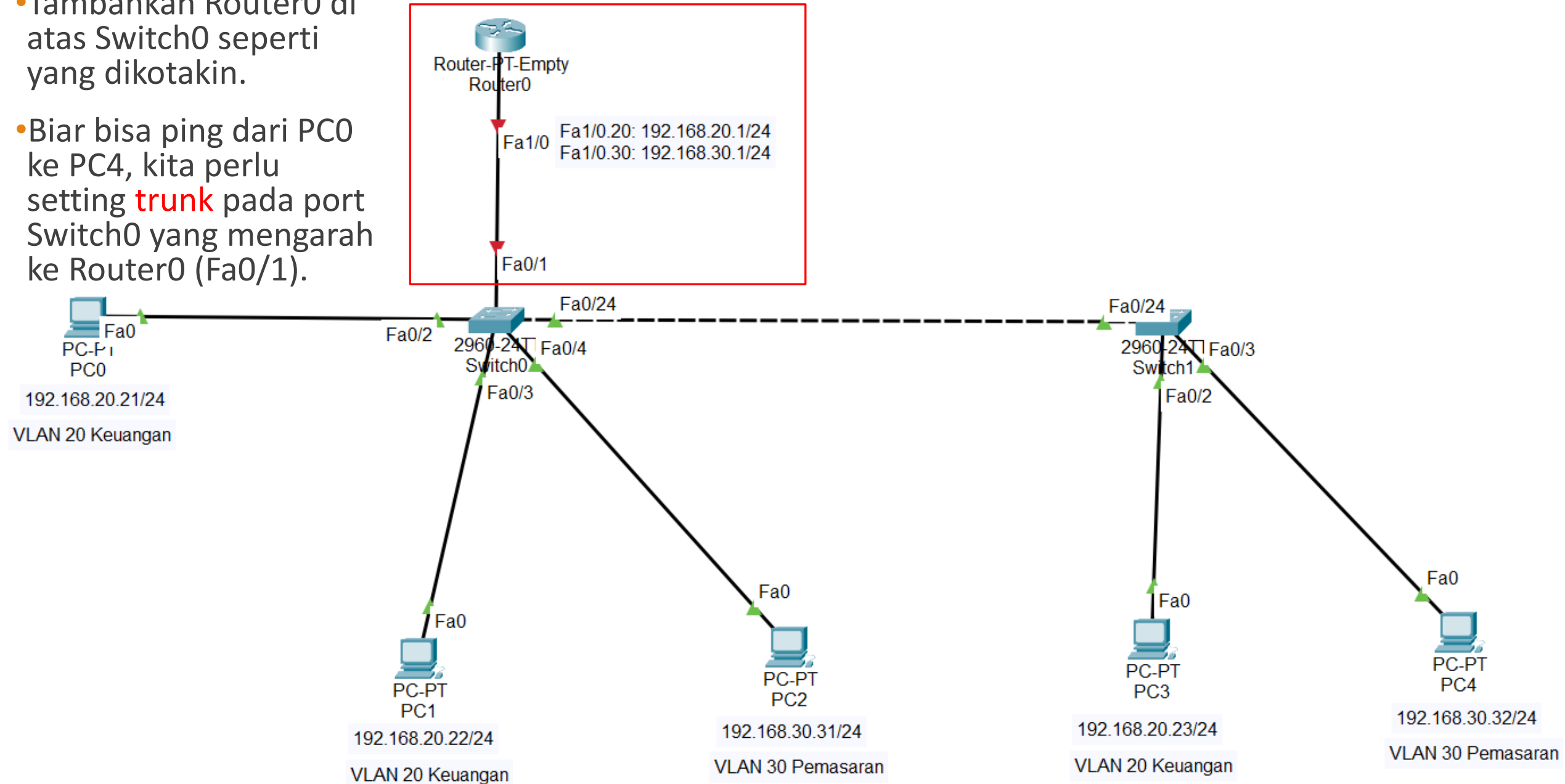
```
Pinging 192.168.30.32 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.32:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

# VLAN dengan Cisco Packet Tracer

- Tambahkan Router0 di atas Switch0 seperti yang dikotakin.
- Biar bisa ping dari PC0 ke PC4, kita perlu setting **trunk** pada port Switch0 yang mengarah ke Router0 (Fa0/1).



# VLAN dengan Cisco Packet Tracer

## Trunk Switch0 ke Router0

---

Konfigurasi perintah berikut pada Switch0:

- Switch>enable
- Switch#configure terminal
- Switch(config)#interface fastEthernet 0/1
- Switch(config-if)#switchport mode **trunk**

# VLAN dengan Cisco Packet Tracer

## Sambungkan Router0 ke Switch0

---

Selanjutnya kita akan melakukan konfigurasi IP pada sub interface Router0 yang mengarah ke Switch0 [ada 2 sub interface (Fa1/0.20 dan Fa1/0.30) untuk 2 VLAN, yaitu VLAN 20 dan VLAN 30].

Konfigurasikan **interface** pada Router0:

- Router>enable
- Router#configure terminal
- Router(config)#interface fastEthernet 1/0 [masuk ke interface Fa1/0]
- Router(config-if)#no shutdown [mengaktifkan (up) interface Fa1/0]
- Router(config-if)#exit

# VLAN dengan Cisco Packet Tracer

## Sambungkan Router0 ke Switch0

---

Konfigurasi **sub interface** pada Router0:

- Router(config)#interface fastEthernet 1/0.20 [masuk ke sub interface Fa1/0.20]
- Router(config-subif)#encapsulation dot1Q 20 [konfigurasi routing pada VLAN id 20 dengan IEEE 802.1Q encapsulation]
- Router(config-subif)#ip address 192.168.20.1 255.255.255.0 [konfigurasi ip pada sub interface Fa1/0.20]
- Router(config-subif)#exit
- Router(config)#interface fastEthernet 1/0.30
- Router(config-subif)#encapsulation dot1Q 30
- Router(config-subif)#ip address 192.168.30.1 255.255.255.0
- Router(config-subif)#exit
- Router(config)#exit

# VLAN dengan Cisco Packet Tracer

## Sambungkan Router0 ke Switch0

---

- Untuk menghapus sub interface, bisa menggunakan perintah:
  - Router(config)#no interface <<nama\_interface>>
  - Contoh: Router(config)#no interface fastEthernet 1/0.20
- Untuk menghapus IP pada interface / sub interface, bisa menggunakan perintah:
  - Router(config-subif)#no ip address
- Periksa konfigurasi ip di Router0 dengan perintah:
  - Router#show ip interface brief

| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0    | unassigned   | YES | unset  | administratively down | down     |
| FastEthernet1/0    | unassigned   | YES | unset  | up                    | up       |
| FastEthernet1/0.20 | 192.168.20.1 | YES | manual | up                    | up       |
| FastEthernet1/0.30 | 192.168.30.1 | YES | manual | up                    | up       |

# VLAN dengan Cisco Packet Tracer

## Sambungkan Router0 ke semua PC

- Konfigurasi default gateway pada semua PC, contoh pada PC0 dan PC2:

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.21

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1 ✓

DNS Server 0.0.0.0

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.30.31

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1 ✓

DNS Server 0.0.0.0

# VLAN dengan Cisco Packet Tracer

## Trunk Switch0 ke Switch1

---

- Coba ping dari PC0 ke PC4, sudah sukses:

```
C:\>ping 192.168.30.32

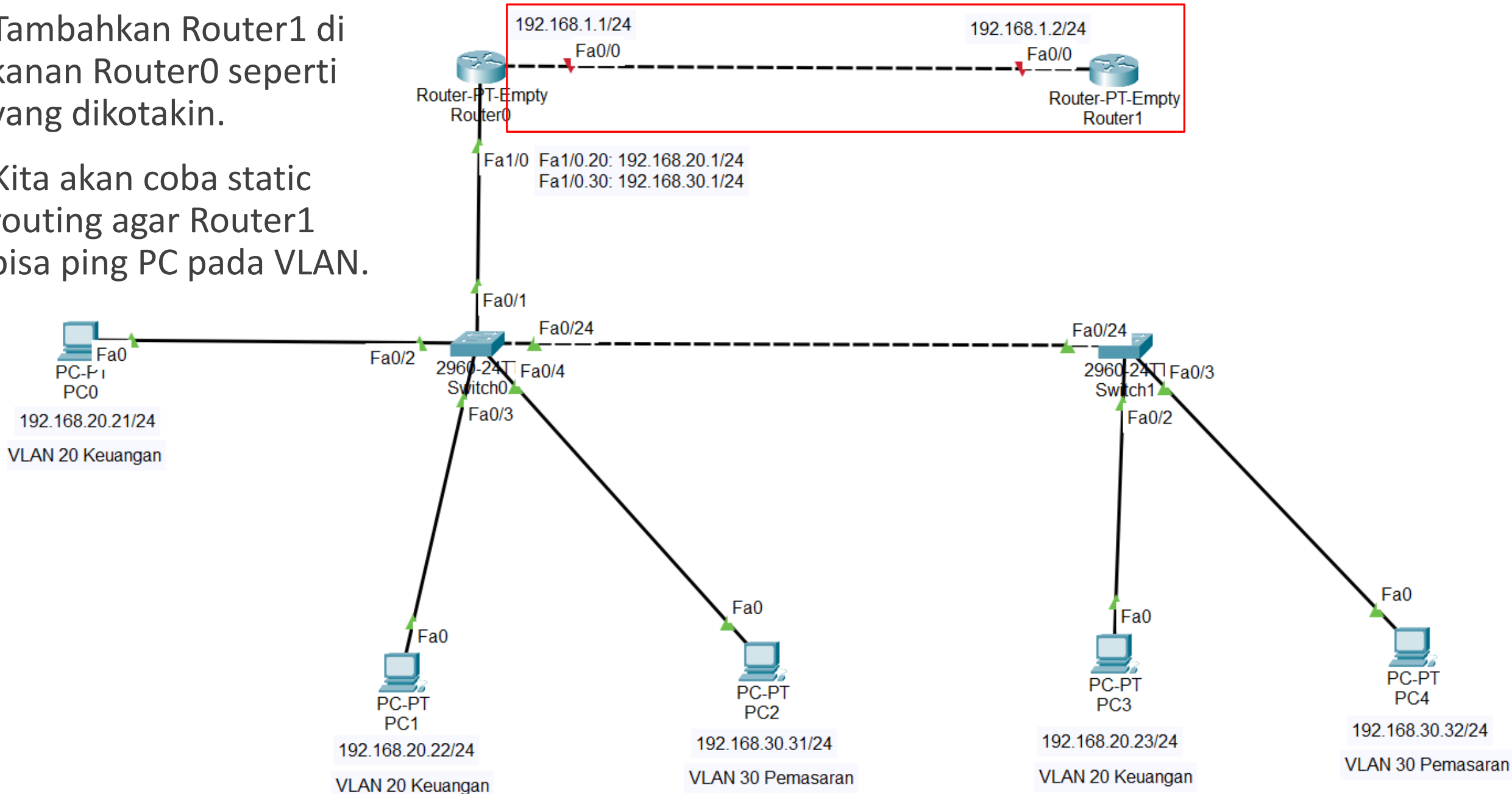
Pinging 192.168.30.32 with 32 bytes of data:

Reply from 192.168.30.32: bytes=32 time<1ms TTL=127
Reply from 192.168.30.32: bytes=32 time<1ms TTL=127
Reply from 192.168.30.32: bytes=32 time=21ms TTL=127
Reply from 192.168.30.32: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.32:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 5ms
```

# VLAN dengan Cisco Packet Tracer

- Tambahkan Router1 di kanan Router0 seperti yang dikotakin.
- Kita akan coba static routing agar Router1 bisa ping PC pada VLAN.



# VLAN dengan Cisco Packet Tracer

## Setting IP Router 0

---

Konfigurasi IP di **interface** Fa0/0 pada Router0:

- Router>enable
- Router#configure terminal
- Router(config)#interface fastEthernet 0/0
- Router(config-if)#ip address 192.168.1.1 255.255.255.0
- Router(config-if)#no shutdown
- Router(config-if)#exit
- Router(config)#exit

# VLAN dengan Cisco Packet Tracer

## Setting IP Router 1

---

Konfigurasi IP di **interface** Fa0/0 pada Router1:

- Router>enable
- Router#configure terminal
- Router(config)#interface fastEthernet 0/0
- Router(config-if)#ip address 192.168.1.2 255.255.255.0
- Router(config-if)#no shutdown
- Router(config-if)#exit
- Router(config)#exit

# VLAN dengan Cisco Packet Tracer

## Setting IP Router 1

---

- Coba ping dari Router1 ke 192.168.1.1, sudah sukses:

```
Router>ping 192.168.1.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:  
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

# VLAN dengan Cisco Packet Tracer

## Setting Static Routing Router 1

---

Konfigurasi static routing pada Router1:

- Router>enable
- Router#configure terminal
- Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.1.1 [routing ke network address 192.168.20.0/24 dengan next hop 192.168.1.1]
- Router(config)#ip route 192.168.30.0 255.255.255.0 192.168.1.1
- Router(config)#exit

# VLAN dengan Cisco Packet Tracer

## Setting Static Routing Router 1

- Untuk menghapus konfigurasi static routing, bisa menggunakan perintah:
  - Router(config)#no ip route <<network\_address>>  
<<subnet\_mask\_desimal\_network\_address>> <<next\_hop>>
  - Contoh: Router(config)#no ip route 192.168.20.0 255.255.255.0 192.168.1.1
- Periksa konfigurasi static routing di Router1 dengan perintah:
  - Router#show ip route

```
Router>show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
```

```
S    192.168.20.0/24 [1/0] via 192.168.1.1
```

```
S    192.168.30.0/24 [1/0] via 192.168.1.1
```

# VLAN dengan Cisco Packet Tracer

## Setting Static Routing Router 1

---

- Coba ping dari Router1 ke PC3, sudah sukses:

```
Router>ping 192.168.20.23
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.20.23, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

- Coba ping dari Router1 ke PC4, sudah sukses:

```
Router>ping 192.168.30.32
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.30.32, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/2 ms
```

# Terima Kasih

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